Harvesting Energy at Your Fingertips:
Harnessing Mechanical Energy from Typing for Sustainable Power Generation

2024

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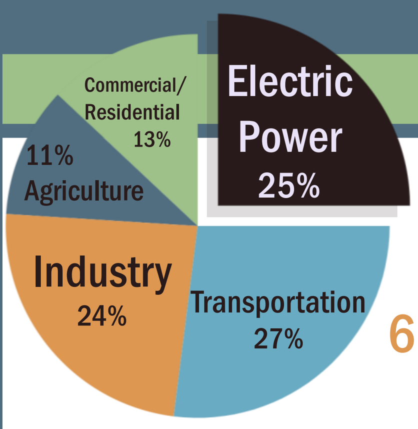
Introduction

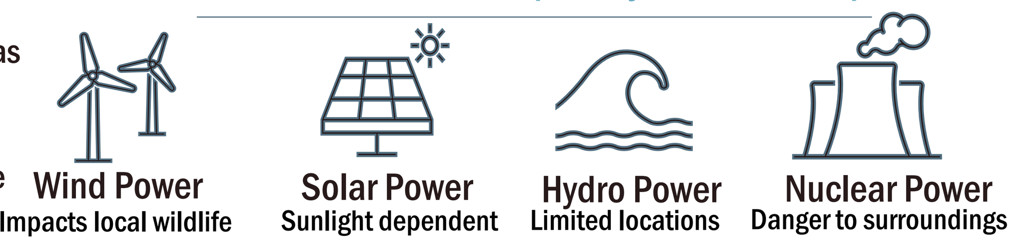
Energy Issues



Electricity production attributes to 1/4 of all green-house gas emissions in the US (EPA.gov, 2020)

61.3% of electricity was derived from burning fossil fuels in 2020 (Kabeyi et al. 2022)

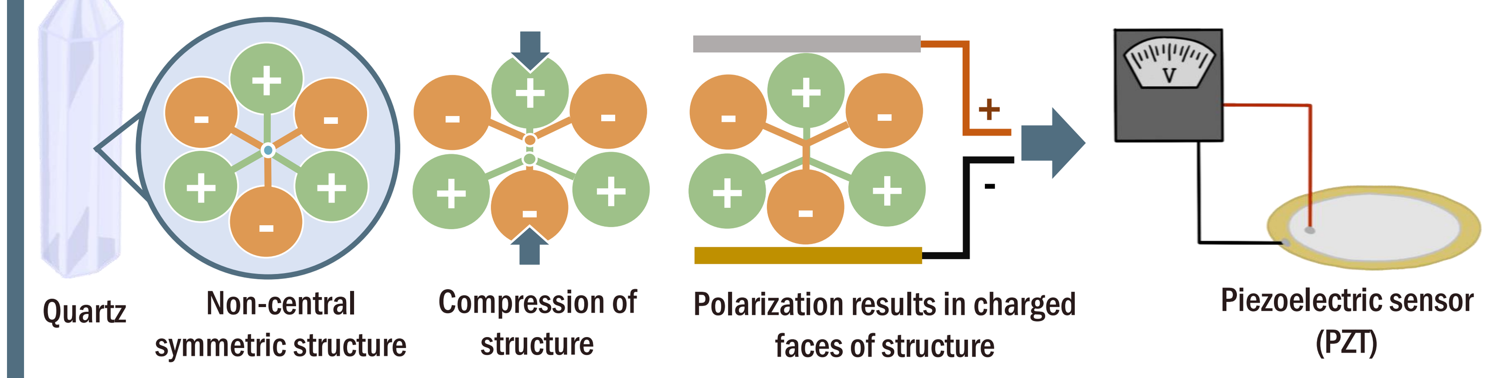
Oil and natural gas reserves are on track to diminish entirely within the next century (BP, 2021)

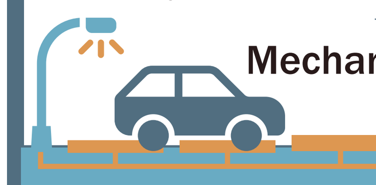


Expanding renewable energy sources is crucial to surpass carbon intensive energy production processes (Sezer et al., 2021)

Piezoelectricity

The phenomenon in which electrical charge is accumulated in certain solids in response to applied mechanical stress (Yan et. al 2021)





Mechanical energy is omnipresent in daily life (Sun et al. 2019)

Moussa et al., 2022 | Xu et al., 2018 | Kumar and Chaturvedi, 2013

Solution

Annual average electricity consumption of approximately 150 kWh per device

Over 2 billion computers exist worldwide today

An average clerical worker presses 1700 keys per hour (Beijing University of Technology, 2012)

Mechanical energy can be converted into electricity, charging a computer with clean energy harvesting practices

Purpose Statement / Gap in Research

The potential of recent piezoelectric advancements for energy harvesting in computers remains unexplored.

Objectives

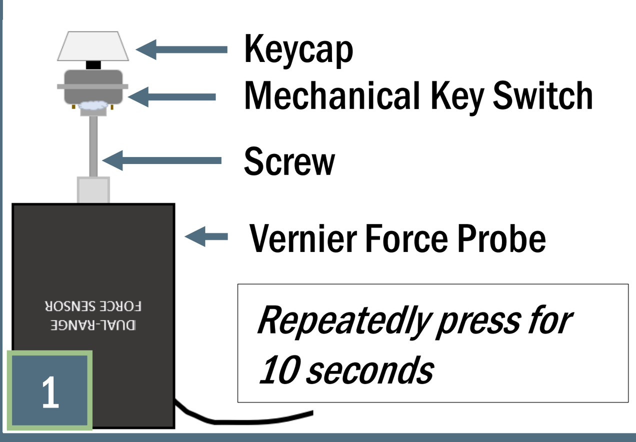
1. Quantify the magnitude of forces on a computer to determine piezoelectric feasibility
2. Compare the potential differences of hard and soft PZT
3. Design and build a piezoelectric computer key that harvests mechanical energy
4. Test piezoelectric potential differnces of constructed computer key

*Student researcher has full ownership over this project; all graphics and photos were made by the student unless otherwise noted

Methods | Results

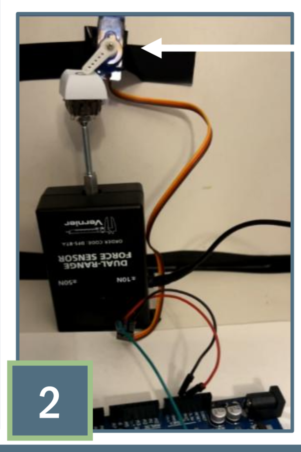
1. Quantify the Magnitude of Force while Typing on a Computer

How much force is applied per keystroke?



Repeatedly press for 10 seconds

How can this force be standardized for testing?

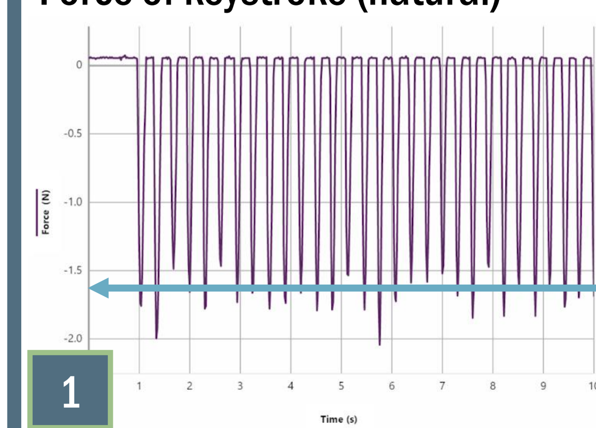


Utilized an SG90 Arduino motor rotating 180°

Coded to apply 1.6N of force to key switch system

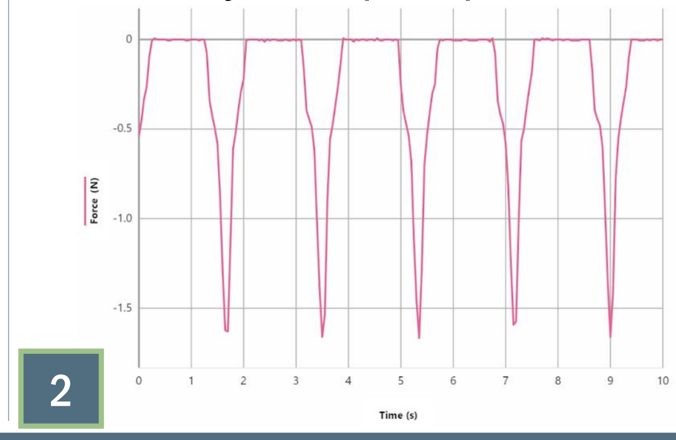
```
#include <Servo.h>


Servo myservo;

int pos = 0;

void setup() {
  myservo.attach(9);
}

void loop() {
  for (pos = 0; pos <= 180; pos += 1) {

    myservo.write(pos);
    delay(5);
  }
  for (pos = 180; pos >= 0; pos -= 1) {

    myservo.write(pos);
    delay(5);
  }
```

Force of keystroke (natural)



Average force of 1.6N per keystroke

Force of keystroke (SG90)



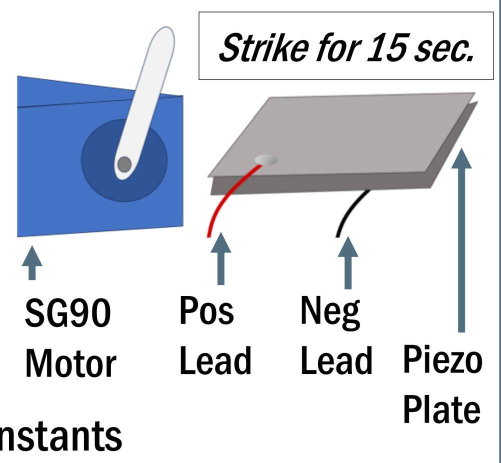
Motor consistently provides the same amount of force to the computer key, standardizing a method for testing

2. Compare the Potential Differences of Soft and Hard PZT

Item 248:	Item 1687:	Item 1377:	Item 1365:	Item 1726:	Item 1738:
Material: Hard Ceramic	Material: Hard Ceramic	Material: Soft Ceramic	Material: Soft Ceramic	Material: Soft Ceramic	Material: Soft Ceramic
Dimensions 10 x 10 mm	Dimensions 10 x 10 mm	Dimensions 10 x 10 mm	Dimensions 14 x 12 mm	Dimensions 11.6 x 10.1 mm	Dimensions 10 x 10 mm
Thickness 1.50 mm	Thickness 0.28 mm	Thickness 0.19 mm	Thickness 1.00 mm	Thickness 0.70 mm	Thickness 0.80 mm

Materials from American Piezo

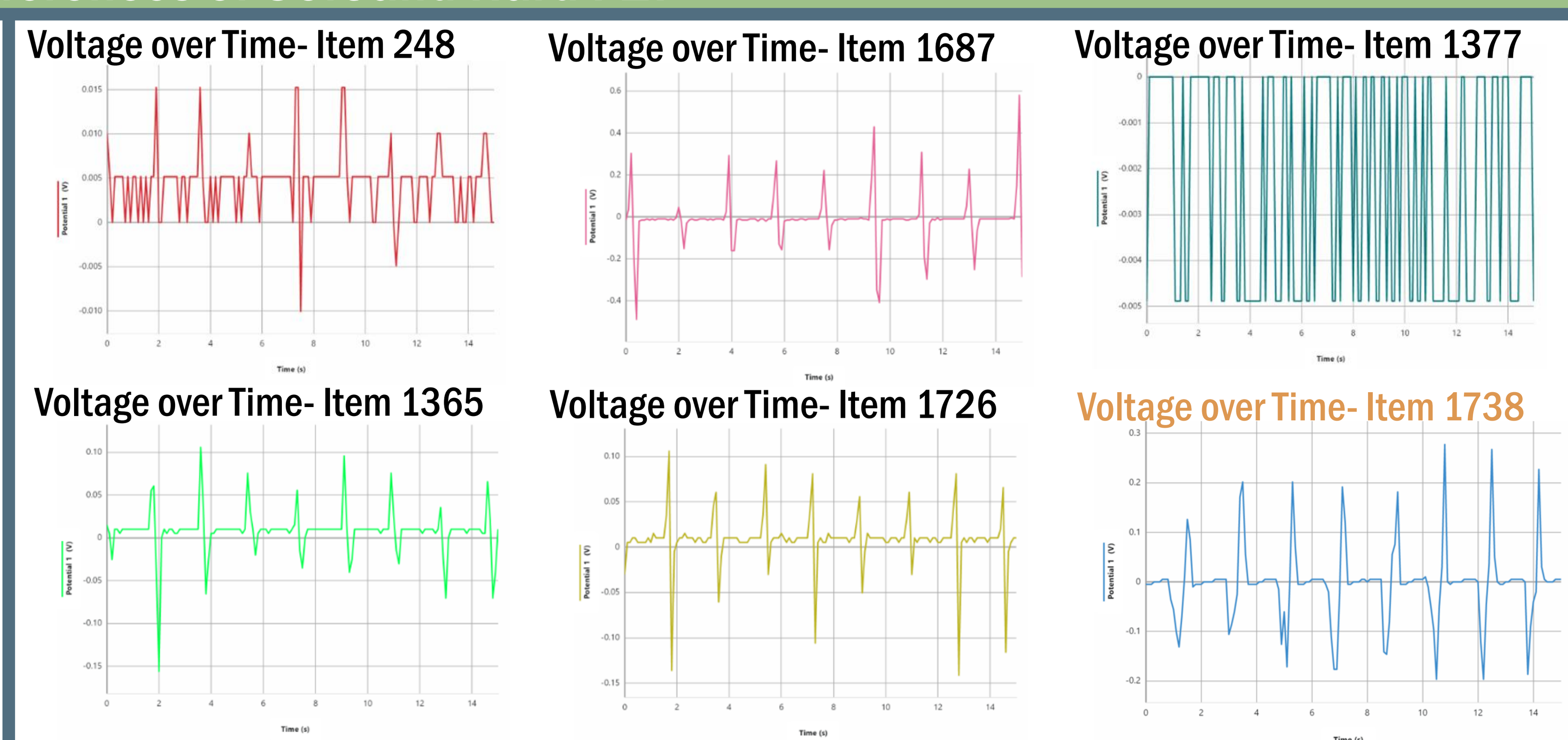
Strike for 15 sec.



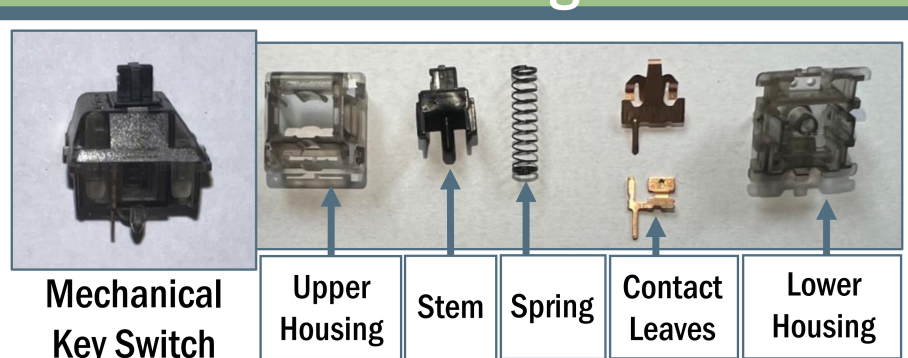
Which type of material would work best in a computer?

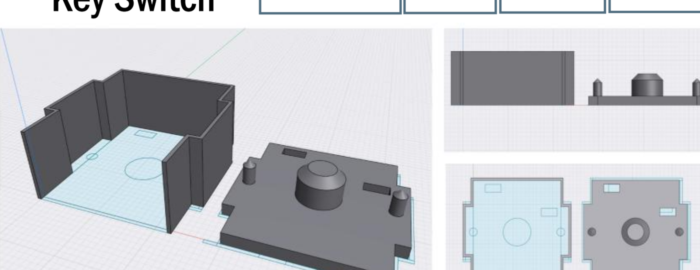
Hard Piezoelectric Ceramic: Resistant to deformation, large dielectric constants

Soft Piezoelectric Ceramic: More flexible and deformable, smaller dielectric constants



3. Design a Piezoelectric Computer Key





3D print new base

Re-attach original elements

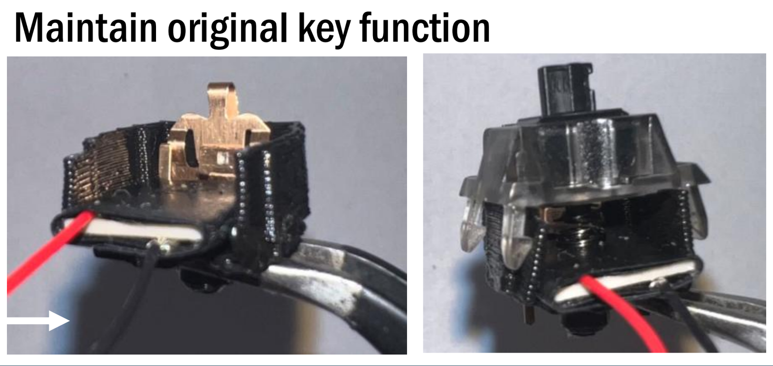
Goals:

Modify base to accommodate Piezoelectric Plate

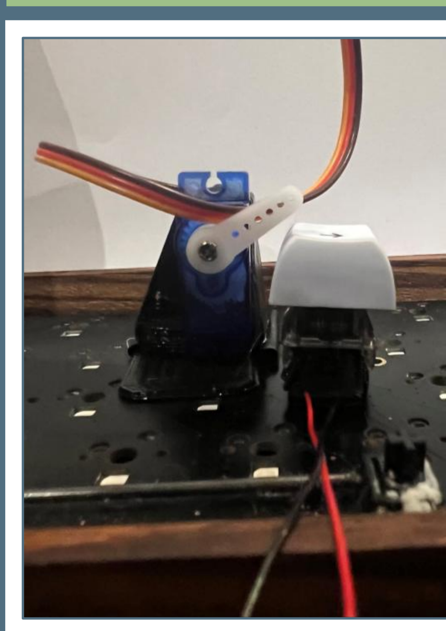
Item 1738- Voltage 0.045V-0.580V

Apply maximum force from stem to Plate

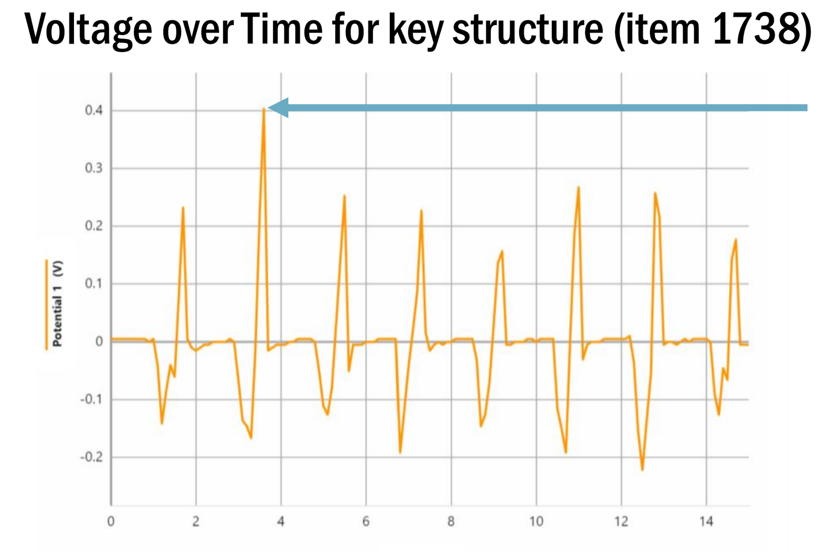
Maintain original key function



4. Test Potential Difference of the Computer Key



Voltage over Time for key structure (item 1738)



0.403 V per keystroke

Computer key can create voltage in response to mechanical stress

Discussion

1. On average, 1.6N of force is applied to a computer key per keystroke
2. There was no significant differnece between the usage of hard or soft ceramic
3. A functional computer key was created
4. The key displayed its ability to convert mechanical energy to electricity

How much *power* is produced while using the device? (P= V *I)

P = 0.403V * 5mA → P = 2.015mW per keystroke

→ 0.003 kWh per hour typing → 0.024 kWh per workday

→ **48,000,000 kWh of electricity worldwide every day**

The average annual electricity consumption for a computer is 146 kWh

The average annual electricity consumption for a home in the United States is 10,632 kWh

Limitations/ Future Work

An efficient self charging computer is a reality

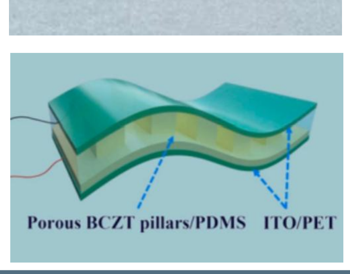
Ceramic piezoelectric resulted in fractures, preventing material usage

Low current leads to a low power output

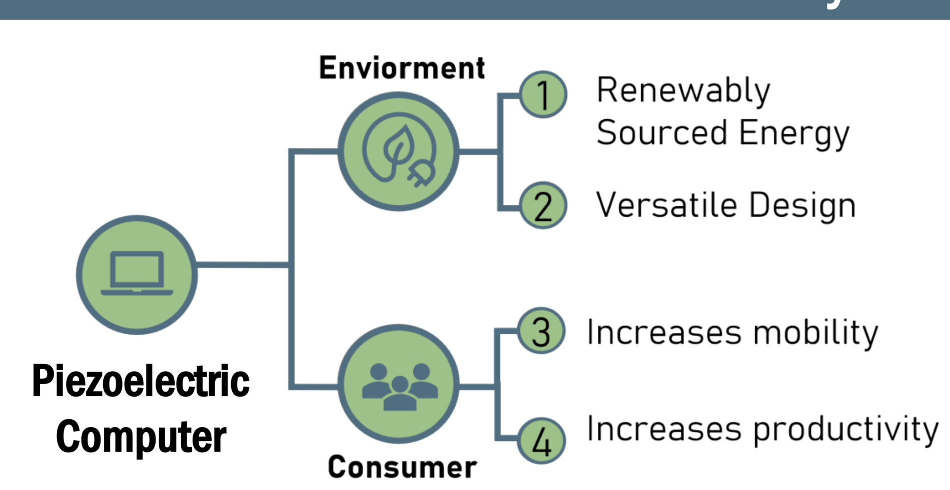
Replacing the piezo elements with high grade piezoelectric transducers will lead to a better produced charge

Yan et al., 2021

Item 1377



Novelty



One of the first autonomously charging devices that does not face vast environmental restrictions

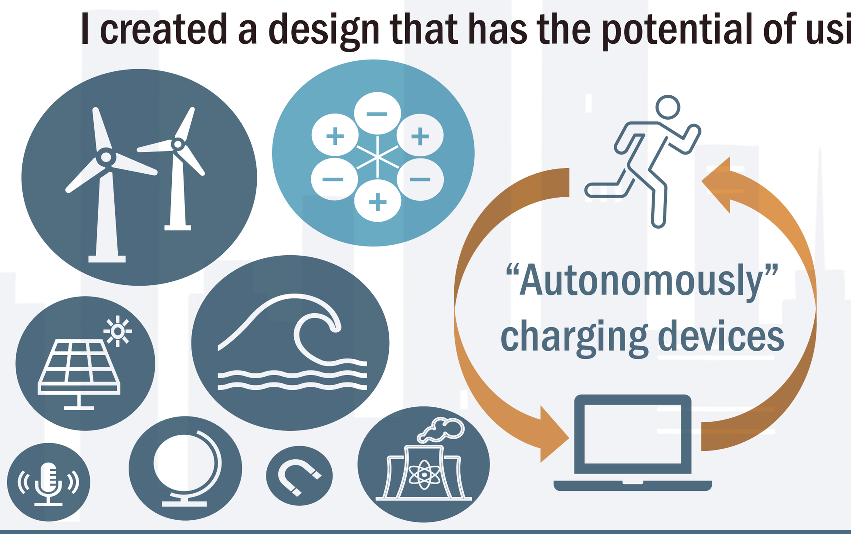
Reduces the non-renewable energy consumption of electronic devices

Removes strain from the electric grid promoting new "green" technologies

Aids in accomplishing Goal 7 (UN)

Conclusion

I created a design that has the potential of using 100% renewable energy (mechanical)



Nearly all piezoelectric transducers are on the rise to make big impacts in the renewable energy field, specifically related to clean energy generation methods, and every-day applications of such devices helps to promote accessible clean-living (Akinga et al. 2020, Chaturvedi et al. 2013, Xu et al. 2017, Petritz et al. 2021)

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